🤖 Convert Your Stock Model into an AI Agent

Complete Guide with 4 Options (Beginner to Advanced)

🎯 What is an Agent?

Your Current Model: You manually run analysis when you want
Agent Model: Runs automatically and alerts you when opportunities arise

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📊 OPTION 1: Simple Scheduled Agent (EASIEST)

What It Does:

- ✅ Runs automatically every day at 9 AM
- ✅ Analyzes your watchlist stocks
- ✅ Sends email if buy opportunities appear
- ✅ No coding needed (just setup)

How to Set Up:

Step 1: Create Alert Rules

Create `agent_config.json`:

```
json
```

```
{
  "watchlist": [
    "stock_config_E2E.json",
    "stock_config_TCS.json",
    "stock_config_RELIANCE.json"
  ],

  "alert_rules": {
    "E2E": {
      "buy_below": 2200,
      "stop_loss": 1900
    },
    "TCS": {
      "buy_below": 3500,
      "stop_loss": 3200
    }
  },

  "email": {
    "enabled": true,
    "sender_email": "your_email@gmail.com",
    "sender_password": "your_app_password",
    "recipient_email": "your_email@gmail.com"
  }
}
```

Step 2: Run the Agent

```powershell
# Run once (test)
py stock_agent_simple.py

# Run every 24 hours continuously
py stock_agent_simple.py --continuous 24
```

Step 3: Schedule with Windows Task Scheduler

1. Open **Task Scheduler** (Windows search)
2. **Create Basic Task**
3. Name: "Stock Agent"
4. Trigger: Daily at 9:00 AM
5. Action: Start Program
    - Program: `C:\Python39\python.exe`

- Arguments: `stock_agent_simple.py`
- Start in: `C:\Users\drdee\AI_Agent\Desktop\StockAnalysis`

6. Done!

Now it runs automatically every day! 📅

🧠 OPTION 2: AI-Powered Agent with Claude (RECOMMENDED)

What It Does:

- ✅ Uses Claude AI to analyze stocks
- ✅ Makes intelligent buy/sell decisions
- ✅ Explains reasoning in plain English
- ✅ Adapts to market conditions

How to Set Up:

Step 1: Get Claude API Key

1. Go to https://console.anthropic.com
2. Sign up / Log in
3. Go to **API Keys**
4. Create new key
5. Copy it

Step 2: Install Anthropic SDK

```powershell
py -m pip install anthropic
```

Step 3: Use the AI Agent

```python
# stock_agent_ai.py (already created for you)
import anthropic

client = anthropic.Anthropic(api_key="your-api-key-here")

# The agent will analyze and give recommendations
```

Benefits:

- 🧠 AI understands market context
- 📊 Learns from multiple data points
- 💬 Explains decisions clearly
- 🎯 More nuanced than rule-based

🖥️ OPTION 3: Web Dashboard Agent (INTERMEDIATE)

What It Does:

- ✅ Beautiful web interface
- ✅ Real-time stock monitoring
- ✅ Interactive charts
- ✅ One-click analysis

Tech Stack:

- Frontend: Streamlit (Python web framework)
- Backend: Your existing models
- Deployment: Streamlit Cloud (free)

Code Example:

```python
```

```
import streamlit as st
from nse_generic_forecaster import NSEStockForecaster

st.title("🤖 AI Stock Analysis Agent")

# Sidebar for stock selection
ticker = st.sidebar.selectbox(
    "Select Stock",
    ["E2E", "TCS", "INFY", "RELIANCE"]
)

# Run analysis
if st.button("Analyze"):
    with st.spinner("Analyzing..."):
        forecaster = NSEStockForecaster(f"stock_config_{ticker}.json")
        result = forecaster.generate_report()

        # Display results
        st.metric("Current Price", f"₹{forecaster.current_price:,.0f}")
        st.metric("Fair Value", f"₹{result['fair_value']:,.0f}")
        st.metric("Quality Score", f"{result['composite_score']:.1f}/10")
```

Install:

```powershell
py -m pip install streamlit
streamlit run stock_dashboard.py
```

Opens at: http://localhost:8501

🚀 OPTION 4: Full Autonomous Trading Agent (ADVANCED)

What It Does:

- ✅ Monitors markets 24/7
- ✅ Automatically places orders via broker API
- ✅ Manages portfolio allocation
- ✅ Rebalances based on signals

⚠️ **WARNING: This trades real money!**

Components Needed:

1. **Broker Integration**
   - Zerodha Kite API
   - Upstox API
   - Alice Blue API
2. **Risk Management**
   - Position sizing (Kelly Criterion)
   - Stop losses
   - Max drawdown limits
3. **Execution Logic**

```python
def execute_trade(ticker, action, quantity):
    """
    action: 'BUY' or 'SELL'
    """
    if action == 'BUY':
        # Check if we have cash
        # Place limit order at support level
        # Monitor for fill
    elif action == 'SELL':
        # Place limit order at resistance
        # Monitor for fill
```

NOT RECOMMENDED unless you're experienced!

🎯 MY RECOMMENDATION FOR YOU

Start with Option 1: Simple Scheduled Agent

Why:

- ✅ Easy to set up (30 minutes)
- ✅ No AI API costs
- ✅ Reliable and predictable
- ✅ You stay in control

Next:

- After 1 month → Add Option 2 (AI analysis)
- After 3 months → Build Option 3 (Dashboard)
- After 6 months → Consider Option 4 (if you want)

📋 STEP-BY-STEP: Build Your First Agent (Option 1)

Complete Setup in 15 Minutes:

1. Download Files (already done ✅)

```
stock_agent_simple.py
agent_config.json (will be created)
```

2. Create Config

```powershell
cd C:\Users\drdee\AI_Agent\Desktop\StockAnalysis
py stock_agent_simple.py
```

This creates `agent_config.json` template.

3. Edit Config

Open `agent_config.json` in Notepad:

```json
{
  "watchlist": [
    "stock_config_E2E.json"
  ],

  "alert_rules": {
    "E2E": {
      "buy_below": 2200
    }
  },

  "email": {
    "enabled": false
  }
}
```

4. Test It

```
powershell

py stock_agent_simple.py
```

You should see:

```
🤖 Stock Agent Running
Analyzing stock_config_E2E.json...
🚨 BUY_OPPORTUNITY: E2E at ₹2,200
✅ Analysis complete. 1 alerts generated.
```

5. Schedule It (Windows)

Option A: Task Scheduler (runs when computer is on)

- See instructions above

Option B: Keep Running (always on)

```
powershell

py stock_agent_simple.py --continuous 24
```

Done! You now have an automated agent! 🎉

🔧 Agent Features Comparison

Feature	Option 1 Simple	Option 2 AI	Option 3 Dashboard	Option 4 Trading
Ease of Setup	⭐⭐⭐⭐⭐	⭐⭐⭐⭐	⭐⭐⭐	⭐⭐
Cost	Free	$0.15/analysis	Free	API fees
Intelligence	Rule-based	AI-powered	Visual	Full auto
Control	Full	High	High	Limited
Risk	None	None	None	HIGH
Best For	Beginners	Everyone	Tech users	Experts

💡 Agent Ideas (Get Creative!)

Idea 1: Sector Rotation Agent

Monitors all IT stocks, recommends switching when one becomes better

Idea 2: News-Triggered Agent

Watches news APIs, alerts when company has major announcement

Idea 3: Technical Pattern Agent

Detects chart patterns (double bottom, breakouts)

Idea 4: Portfolio Rebalance Agent

Monitors your holdings, suggests rebalancing quarterly

Idea 5: Dividend Calendar Agent

Tracks ex-dates, alerts for dividend capture opportunities

🆘 Troubleshooting

Agent not sending emails?

For Gmail:

1. Enable 2-Factor Authentication
2. Create App Password:
    - Google Account → Security → App Passwords
    - Select "Mail" and generate
    - Use this password in config (not your main password)

Agent crashes?

Add error handling:

```python
try:
    agent.run_daily_analysis()
except Exception as e:
    print(f"Error: {e}")
    # Send error email to yourself
```

Agent using too much CPU?

Reduce frequency:

```powershell
# Run every 6 hours instead of continuous
py stock_agent_simple.py --continuous 6
```

📚 Learning Resources

To Build Better Agents:

1. **Python Automation:**
   - Book: "Automate the Boring Stuff with Python"
   - Free: https://automatetheboringstuff.com
2. **AI Integration:**
   - Anthropic API Docs: https://docs.anthropic.com
   - Claude Cookbook: https://github.com/anthropics/anthropic-cookbook
3. **Trading APIs:**
   - Zerodha Kite: https://kite.trade/docs/connect/v3/
   - Upstox: https://upstox.com/developer/api-documentation/
4. **Web Dashboards:**
   - Streamlit Docs: https://docs.streamlit.io
   - Tutorial: https://www.youtube.com/watch?v=JwSS70SZdyM

🎯 Your Next Steps

Week 1:

- ✅ Set up simple agent (Option 1)
- ✅ Test with 1-2 stocks
- ✅ Get comfortable with alerts

Week 2:

- ✅ Add all your watchlist stocks
- ✅ Fine-tune alert rules
- ✅ Set up email notifications

Week 3:

- ✅ Get Claude API key
- ✅ Try AI-powered analysis (Option 2)
- ✅ Compare AI vs rule-based

Month 2:

- ✅ Build simple dashboard (Option 3)
- ✅ Track accuracy over time
- ✅ Refine your strategy

⚠️ Important Warnings

DO:

- ✅ Start simple (Option 1)
- ✅ Test thoroughly before relying on it
- ✅ Keep human oversight
- ✅ Review agent decisions weekly

DON'T:

- ❌ Let agent trade automatically (yet)
- ❌ Ignore stop losses
- ❌ Over-optimize on past data
- ❌ Risk more than you can afford to lose

🎓 Final Advice

Remember:

1. **Agent is a tool, not a replacement for judgment**
2. **Start conservative, scale up gradually**
3. **Always verify agent recommendations**
4. **Keep learning and improving**

An agent that saves you 30 minutes/day = 180 hours/year!

That's worth building! 🚀

Ready to build your agent? Start with Option 1 now!